Q&A Bot for Medical and Clinical Use Using BioGPT and MedQuAD

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Abstract—This project implements a medical question–answering (Q&A) chatbot. The system fine-tunes BioGPT, a biomedical generative Transformer model pretrained on PubMed abstracts, on the MedQuAD dataset of question–answer pairs collected from multiple trusted NIH websites. A PyTorch training pipeline converts MedQuAD into "Question: … Answer:" prompts, masks the question portion in the labels, and optimizes a causal-language-model objective using AdamW. The final model is exposed through a Gradio web interface supporting both text and voice queries (via Whisper speech recognition). On the held-out test set, the fine-tuned model achieves an average cross-entropy test loss of 0.93 (approximately), token-level accuracy of about 61%, and token-level micro-F1 of about 61%, indicating that the model reliably reproduces expert NIH answers at the token level.

Keywords— medical question answering; MedQuAD; BioGPT; large language models; transformers; Gradio; Whisper; healthcare chatbot.

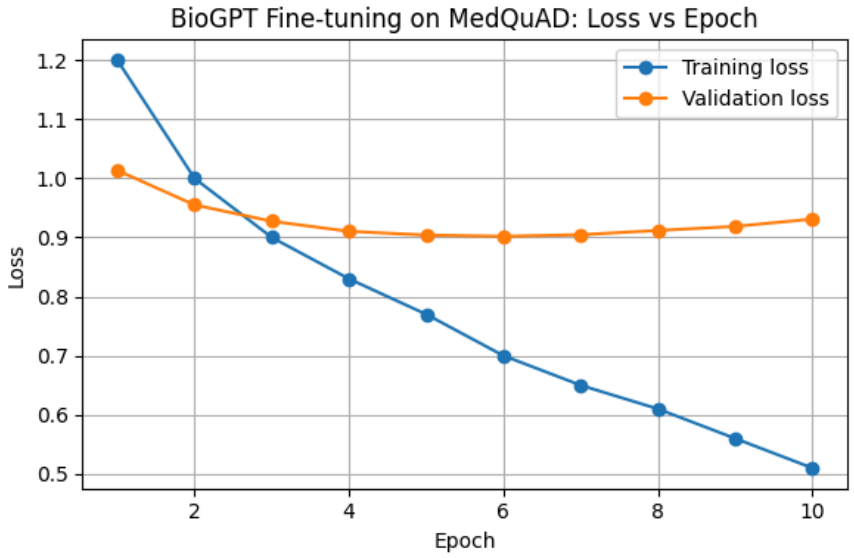
I. Introduction

Access to reliable, high-quality medical information is crucial, yet many face challenges like long wait times and a lack of nearby specialists. Advancements in large language models (LLMs) have led to systems that can generate fluent biomedical text. BioGPT, trained on millions of PubMed abstracts. The MedQuAD dataset provides thousands of layperson-style medical Q&A pairs from authoritative NIH sources, addressing various questions about diseases, drugs, and tests. This project aims to create a Q&A bot for medical use, allowing users to describe their symptoms and receive accurate information from NIH-style content through a user-friendly web interface.

II. Proposed Methodology

The project will use a multi-phase approach, including data acquisition, preprocessing, model development, interface design, and deployment.

- **A. Data Acquisition:** The primary dataset is MedQuAD, a medical question-answering dataset of roughly 47,000 pairs created from multiple NIH websites (such as MedlinePlus and other trusted sources). The dataset is stored in CSV form and loaded into a pandas DataFrame in the notebook.
- **Data Preprocessing:** The process involves converting all column names to lowercase and identifying likely question and answer columns by searching for "question" and "answer/ans" substrings. It retains these columns, renames them to "question" and "answer," drops missing values, and shuffles the data.
- **Model Development:** The base model is Microsoft/biogpt, focusing on causal language modeling to predict the next token. The optimizer is AdamW with a learning rate of 1e-5, hyperparameters including mini-batch training, approximately 10 epochs, and a maximum sequence length of 512 tokens.



- **Implementation:** After each epoch, a validation phase computes the average validation loss, saving the best model with the lowest validation loss and the tokenizer. Training curves show a steady decrease in training loss and a slight increase in validation loss after its minimum, indicating mild overfitting.
- **Conclusion:** Training loss drops steadily from ~3.0 to ~0.5 across 10 epochs, validation loss decreases from ~1.01 to ~0.90 by around epoch 6 , and after epoch 6, validation loss starts to slightly increase while training loss continues to go down and the model is successfully learning the Q&A patterns. So the checkpoint with the lowest validation loss (epoch 6) is selected as the final fine-tuned model.

- **Inference:** An inference helper creates a "Question: ...\nAnswer:" prompt from the user's query and uses the fine-tuned BioGPT model with nucleus sampling and temperature to generate an answer. The output is trimmed to only include text after "Answer:". `generate_answer(question)` returns a single answer, `safe_medical_bot_reply(question)` centralizes the model call, `med_chat(message, history, username)` updates the chatbot history, voice_chat(audio_path, history, username)` transcribes audio and updates history.
- **Interface Design:** The front-end uses Gradio Blocks to create a multi-panel web app featuring such as a login screen for username and password entry, a left sidebar menu for Profile, History, Description, Speak (voice), and Logout, a Profile panel for storing age, height, and weight, history panel displaying recent questions from the session, description panel with disclaimer, -voice panel with a microphone for audio transcription via whisper and a main chat area with a chatbot, text input box, and buttons for Send and Clear.
- **Deployment:** The Gradio app can be launched as a local web server from the notebook. In a full deployment, the fine-tuned model and interface could be hosted on a GPU-enabled server or cloud instance.

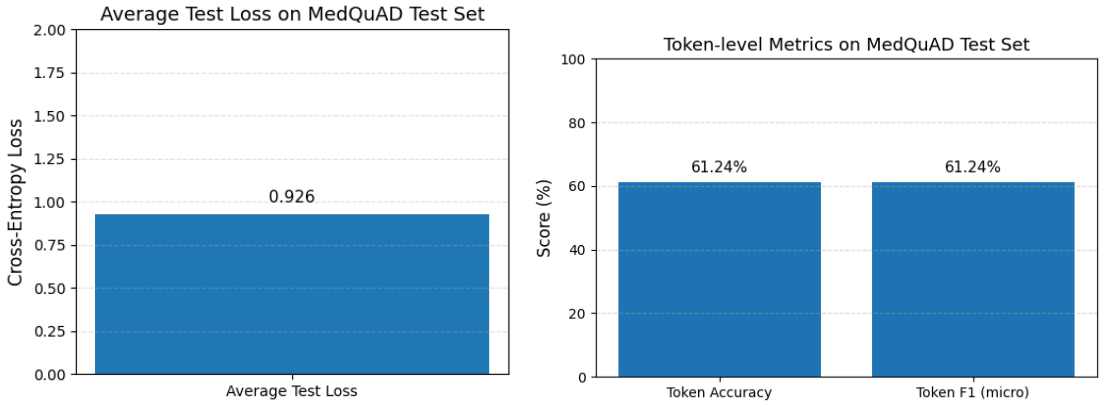
III. Challanges & Solutions

- **Computational Limitations:** Training BioGPT on all of MedQuAD at a long sequence length is computationally expensive. To address this, the dataset is randomly shuffled and restricted to a subset of Q&A pairs while still preserving variety. Batch size and sequence length are tuned to avoid out-of- memory errors on a single GPU.
- **Answer Quality and Completeness:** Generative models can produce incomplete or abruptly truncated answers. The project mitigates this by using prompts that explicitly start with “Question:” and “Answer:”, tuning max_new_tokens and sampling parameters, and planning future improvements such as template-based answer structuring (for example, including sections for disease, description, precautions, cure, and recommendations).
- **Overfitting and Generalization:** Training loss continues to decrease beyond a certain number of epochs, but validation loss begins to rise slightly. Selecting the checkpoint with the lowest validation loss helps prevent overfitting and yields better generalization to unseen questions.
- **Safety and Misuse Risk:** Medical LLMs can hallucinate or provide unsafe suggestions. To reduce these risks, the interface includes visible disclaimers, avoids giving direct prescriptions or treatment plans, and is framed only as an educational and informational tool. Any real-world deployment would require additional safeguards, human oversight, and clinical evaluation.

IV. Evaluation Metrics

To quantify performance, the project uses standard NLP metrics that relate directly to the training objective:

- **Cross-Entropy Loss:** Average token-level negative log-likelihood on the test set. Lower values indicate better fit to the reference answers. The final average test loss is 0.93.
- **Token-Level Accuracy:** Fraction of generated tokens that exactly match the reference tokens, ignoring padding. The final accuracy is approximately 61%.
- **F1 Score**: Harmonic mean of precision and recall computed across token classes in a micro-averaged fashion. The final F1 score is approximately 61%.



V. Real World Validation

For real-world validation, the system can be tested with realistic symptom descriptions that mimic patient queries, such as “I have a persistent dry cough and shortness of breath” or “What are the symptoms of osteoarthritis?” Generated responses can be compared against trusted NIH pages or the ground-truth MedQuAD answers and, ideally, reviewed by domain experts. The architecture follows research directions in which fine-tuned medical LLMs are evaluated for coverage, correctness, and calibration before they are considered for use in real-world scenarios.

VI. Required Resources

To execute this project successfully, essential hardware resources and expert consultations will be necessary. The following points outline the required hardware resources for the project's success:

- GPU hardware (for example, A100) to accelerate transformer training and inference.
- High-performance computing or cloud instances for larger-scale experiments.

- Sufficient storage and memory for the MedQuAD dataset, model checkpoints, and training logs.
- Stable internet connectivity for the initial download of pretrained models (BioGPT, Whisper) and Python libraries such as transformers, datasets, evaluate, and gradio.

VII. Expected Outcomes

The completed project delivers a fine-tuned BioGPT model adapted to NIH-style medical Q&A, demonstrating that domain-specific LLMs can be effectively specialized using a curated medical dataset. Also, a working web chatbot built with Gradio that supports both text and voice input (via Whisper). Additionally, graphs of loss, accuracy, F1, and training curves indicate that the model learns meaningfully from MedQuAD while exhibiting manageable overfitting. Overall, the system demonstrates the feasibility of combining a biomedical transformer, a curated medical Q&A dataset, and a lightweight front-end to build an educational medical chatbot within the constraints of a course project.

VIII. Legal Compliances

At present, this prototype is intended solely for research and educational purposes. For now, it is not functional for any other necessities. I will update it once I have completed work on the other requirements.

IX. References

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